

ImageLab Progress Summary Report

Project: ImageLab

Reporting Period: Week 1

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Week 1 of the ImageLab project focused on building the foundation for a secure, reliable image upload system and preparing the application for async image processing in future development phases. The primary objective was to establish the database structure, implement the image-upload endpoint, and create the frontend workflow to support image submissions.

The database layer was completed with three migrations that created the images, jobs, and image_variants tables. Although only the images table is currently used, the full data model is now in place to support upcoming async processing and image variant generation. The local PostgreSQL environment was configured successfully, including the creation of the required database role, assignment of schema permissions, and successful execution of all migrations.

The image upload API was implemented through the POST /v1/images endpoint with a strong emphasis on security and data integrity. Uploaded files are validated using server-side MIME type detection rather than client-provided file extensions or content types. A maximum upload size of 10 MB is enforced via server-side controls, and each file is assigned a randomly generated filename before being stored.

On the frontend, the upload workflow was completed across the application's modules. Selecting a file generates only a local preview and does not immediately communicate with the server. Network requests occur only when the user explicitly submits the upload. Additional safeguards were introduced to prevent duplicate submissions by immediately disabling the submit button and setting an internal submission flag before async processing begins.

User interface improvements were made to support responsive layouts across different screen sizes. Fluid spacing was implemented using CSS clamp() functions, an additional tablet breakpoint was introduced, and layout adjustments were made to ensure the application remains centered and visually balanced on larger displays.

Overall, Week 1 successfully created the application's basic skeleton and upload infrastructure. With these foundations completed, the project is now ready to begin implementing async job processing, image variant generation, and worker functionality during Week 2.